General Mathematics — Exercise Sheet 24

Integration techniques

Variant: Statements

Exercise 1 (Cyclic parts with two real parameters). For every $a, b \in \mathbb{R}$, find all primitives of $e^{ax}\sin(bx)$ on $\mathbb{R}$. Use the two product derivative identities, as in Lecture 24, to construct a primitive of the form $e^{ax}(A\sin(bx) + B\cos(bx))$ when possible. Handle $a = 0$, $b = 0$ and $a = b = 0$ explicitly, without dividing by a parameter that may vanish. Verify the formula by differentiation.

Exercise 2 (A shifted repeated quadratic). Use partial fractions and the repeated-quadratic model of Lecture 24 to find every primitive on $\mathbb{R}$ of

$$\frac{2x^3 - 3x^2 + 8x - 3}{(x^2 - 2x + 5)^2}.$$

State why no real point is excluded. Show the variable change and scale factor when applying J_2 from that lecture.

Exercise 3 (A cancelled factor and the original domain). Find every primitive of

$$f(x) = \frac{x^3 + 2x^2 - x - 2}{(x-1)(x-2)} \quad \text{on } D = \mathbb{R} \setminus \{1, 2\}.$$

Find the finite limit of the integrand at $x = 1$ and decide whether there is a finite limit at $x = 2$. Determine which members of the primitive family extend continuously through 1, whether those extensions are differentiable there, and whether any member extends continuously through 2. Keep the original domain after cancelling a factor.

Exercise 4 (A trigonometric quotient). Find every primitive of $1/(\sin x \cos x)$ on its real domain. Use $u = \tan x$ and state its intervals between successive excluded points and their independent integration constants. Verify the derivative on each interval.

Exercise 5 (An algebraic square-root substitution). Find every primitive of $1/(x + \sqrt{x})$ on its real domain, then select F satisfying $F(1) = 0$. Use the positive square-root substitution and justify the cancellation that it permits. Verify the result in the original variable.

Exercise 6 (An Euler substitution for a quadratic square root). Find every primitive of $1/\sqrt{x^2+1}$ on $\mathbb{R}$ using the Euler substitution $u = x + \sqrt{x^2+1}$. Show that $u > 0$ for every real x, and derive formulas for x, $\sqrt{x^2+1}$ and $\mathrm{d}x$ in terms of u. Use the positive square-root branch throughout and verify the final primitive.

Exercise 7 (Half-angle branches and a global primitive (⋆)). Let $f(x) = 1/(2+\sin x)$ for $x \in \mathbb{R}$ and set $I_k = (-\pi + 2k\pi, \pi + 2k\pi)$ for $k \in \mathbb{Z}$.

1. On I_0, use $t = \tan(x/2)$ to derive

$$\sin x = \frac{2t}{1+t^2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad \mathrm{d}x = \frac{2\,\mathrm{d}t}{1+t^2}.$$

   Find a primitive of f there.
2. Extend the formula to one primitive on all of $\mathbb{R}$ by choosing the constants on the intervals I_k and assigning values at their boundary points $(2k+1)\pi$. Prove differentiability at those points using Lecture 9's mean value theorem, then state the complete family of global primitives.

The upper line below marks boundaries of the substitution intervals. The lower line shows that they are included in the original domain.

